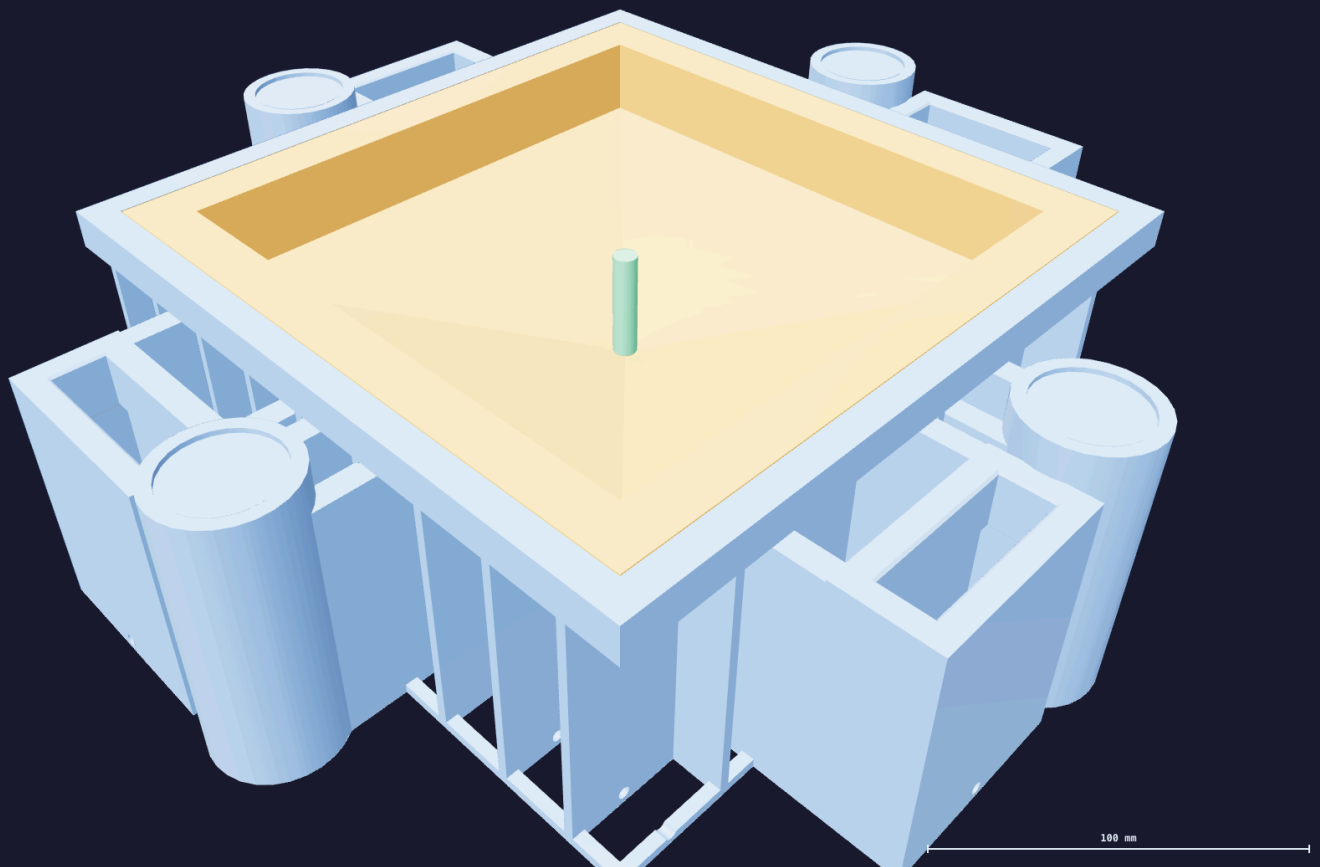


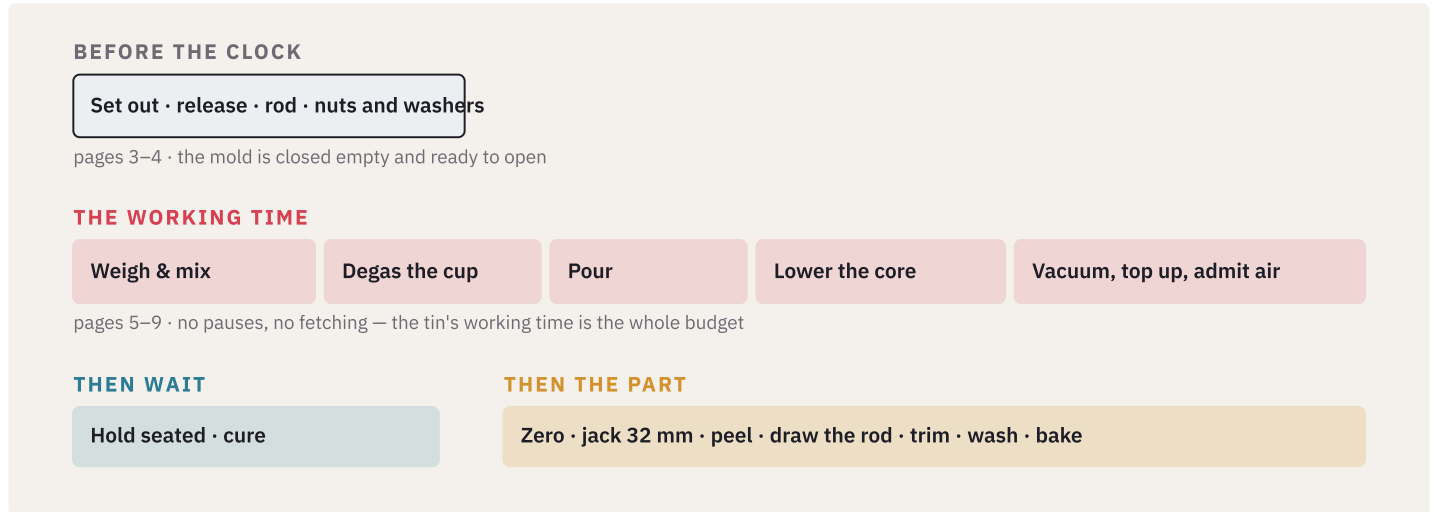
Casting a funnel

Weigh, mix, degas, pour, close, vacuum, cure, jack, peel, trim, bake. One part per run, about 153 g of platinum silicone, and a working time you do not control.



The whole cast

Eleven moves. Five of them happen inside one working time and cannot be paused, so everything they need is laid out before the tins are opened.



The red band is the only part of this you cannot stop in the middle of.

⚠ THE NUMBERS THIS GUIDE DOES NOT HAVE

Working time, cure time and post-cure schedule are on the silicone's own datasheet, not in this machine's files. Read the tin before you start and write its numbers in the margin here. Figures for those quoted elsewhere in the project are planning estimates.

Set the bench

Everything below is in reach and ready **before** either tin is opened. Once the two parts are together you cannot leave the bench.

ON THE BENCH

- ☐ **The mold, coated and dry**, cavity open and level, core beside it.
- ☐ **The ground rod**, clean, checked to bottom at **28.8 mm**.
- ☐ **Four square nuts** in the core arms, retained by their screws, screws **backed out** so the tips clear the washers.
- ☐ **Four washers** flat in the cavity pockets.
- ☐ **The holding arrangement** for the core — the board and weight you settled on the dry run.
- ☐ **A catch tray** under the mold, wider than **270 mm**, for overflow and foam.
- ☐ **Release agent** — Ease Release 200.
- ☐ **Scale** reading to **0.1 g**, one **1000 mL** mixing cup, stir stick, nitrile gloves.
- ☐ **Silicone A and B**, and the black pigment.

IN THE CHAMBER

- ☐ Chamber clear, gauge readable, shutoff valve to hand, pump warm and connected.
- ☐ A flat shelf the cavity can stand on with its **backing-air exits still open**.
- ☐ Room, with the lid closed, for the mold *and* the tray *and* the holding board — you proved this on the dry run.

⚠ GLOVES, AND ONE PUMP AT A TIME

Nitrile gloves for both the release and the silicone. The vacuum pump is shared with the refrigerant work — make sure it is not mid-evacuation somewhere else before you mix.

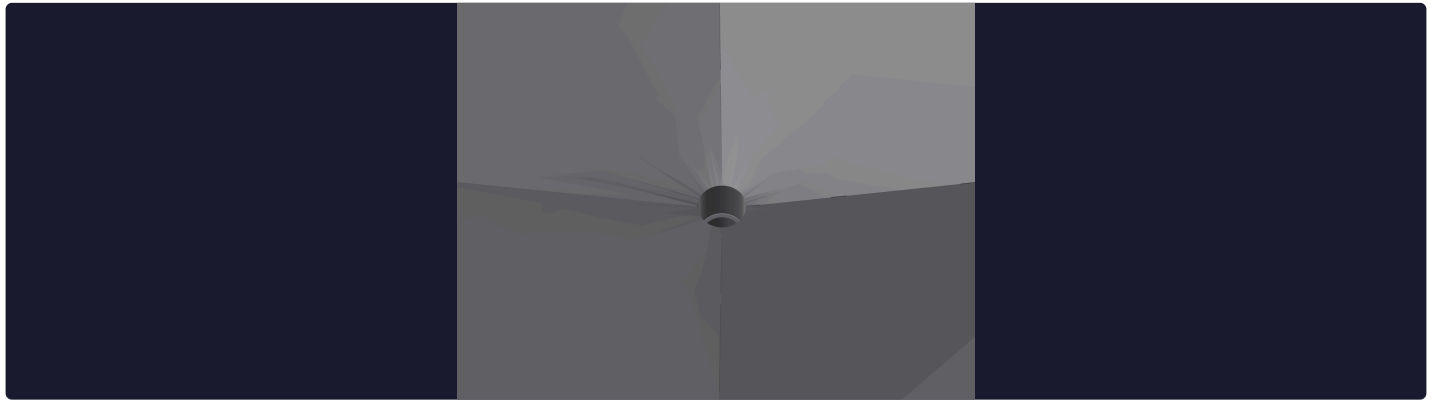
DONE WHEN

Every box above is ticked and you could complete the pour without walking away from the bench.

Release, then the rod

Two faces get release, every single pour. It is a film, not a lubricant, and it wears — a thin patch is where a spout wall tears.

- 1 **Release the cavity's bowl.** Even coverage right into the spout pocket and up the corners. Let it flash off.
- 2 **Release the core's plug** and the brim recess in the plate's underside — the whole of the coated face.
- 3 **Keep it off everything else.** Nuts, guides, washer seats, parting lands, and the inside of the rod socket. Overspray in a guide is a bound guide.
- 4 **Seat the rod in the core** last, by hand, until it bottoms. **22.0 mm** stands out below the plug.



Looking down into the cavity's spout pocket — the deepest, narrowest part of the mold, and the one the rod comes down into.

WHERE THE ROD ENDS UP

With the core seated, the rod's tip stops **2 mm** above the pocket floor. That 2 mm of silicone closes over the end — it is why the demolded spout looks blind, and why you cannot judge the bore before you trim.

DONE WHEN

Both forming faces carry an even release film, nothing else does, and the rod is bottomed in a dry socket.

Weigh and mix

The clock starts when A and B meet. From here to the closed, degassed, topped-up mold is one continuous run.

WHAT	HOW MUCH	WHY THAT
Part A	76.5 g	1 : 1 by weight , not by volume. 153 g is the cast including its sacrificial tip.
Part B	76.5 g	
Black pigment	≤ 2 %, about 1.5 g	Platinum-cure pigment only. Black hides concentrate staining in service.
Mixing cup	1000 mL	Sized for the rise in the chamber, not the pour — expect 3–4×.

- 1

Tare, weigh A, tare, weigh B. To 0.1 g . A platinum system that is off ratio does not cure soft — it cures wrong, or not at all, in patches.
- 2

Pigment into one part first and disperse it there before the two meet. Chasing pigment streaks in the mixed batch spends working time you need later.
- 3

Mix slowly and completely. Scrape the cup wall and the bottom corner twice. Unmixed material on the wall is the tacky patch you find on the demolded funnel.
- 4

Do not hurry this to save air. The chamber removes the air. It cannot remove an unmixed streak.

⚠

KEEP THE CURE CLEAN

Platinum silicone is fussy about what it touches before it gels. Fresh cup, fresh stick, clean gloves, and keep sulphur-bearing rubber, tin-cure silicone and solvent rags away from the bench.

DONE WHEN

One uniform, single-coloured batch of about 153 g, with no streaks on the cup wall.

Degas in the cup

Degas the mixed silicone **in its own container**, before it goes anywhere near the mold. It needs room to climb that the mold does not have.



There is no time on this. The silicone rises, crests and collapses; you pull it **until it stops rising**, and that is the whole instruction.

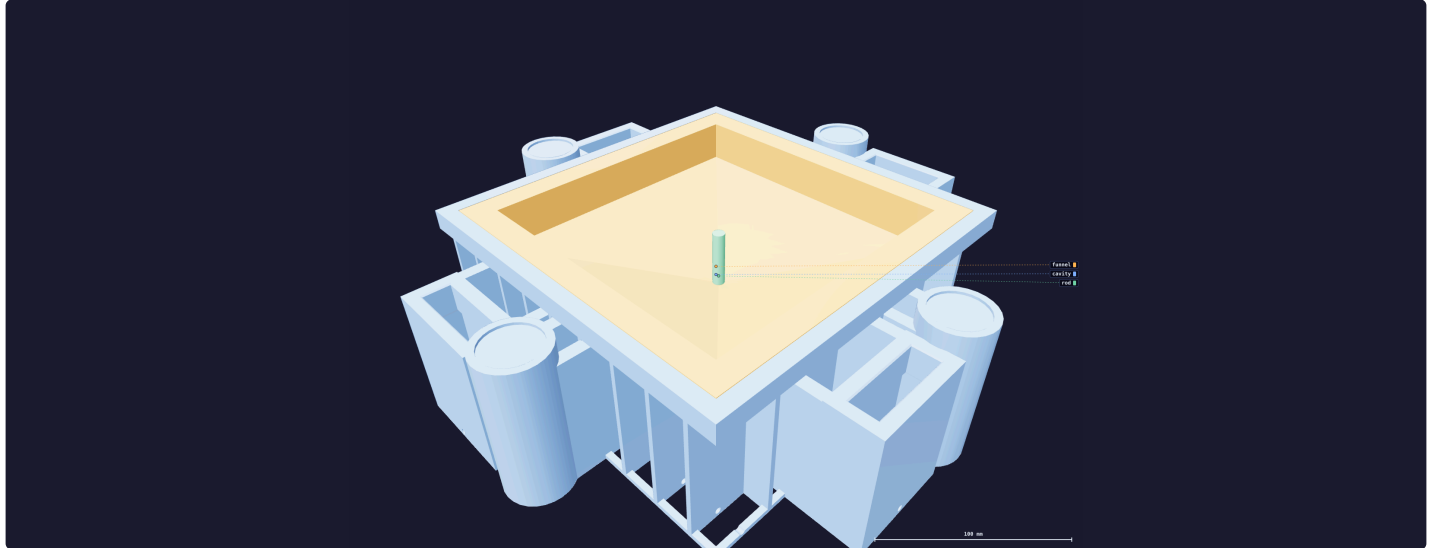
- 1 **Cup into the chamber, lid on, pump on.** Watch it through the glass.
- 2 **If it climbs toward the rim, bleed a little air** to knock it back, then pull again. Silicone over the rim is silicone in your chamber.
- 3 **Hold until the surface is flat and stays flat**, then admit air gently and take it out.

DONE WHEN

The mix has risen, collapsed, and stopped moving under vacuum.

Pour the open cavity

The mold is **open** for the pour. The silicone goes into the bowl and the level rises from the deepest point — it does not fall down a shaft. The core comes down onto it afterwards.



Where it ends up: silicone in the cavity, the rod standing in the spout. The pour is a puddle that grows, and the deep blind spout pocket is the first thing it fills.

- 1 Pour a thin, unbroken stream into the middle of the bowl**, onto the sloping floor rather than straight down the spout. Keep the stream narrow and close — a long fall folds air in.
- 2 Let the spout fill from below.** Watch the level creep up out of the pocket. If a bubble is trapped down there, now is when you can still tease it out with a stick; after the core is down you cannot reach it.
- 3 Put the whole batch in.** The bowl holds far more than the cast needs — the level will look alarmingly low. It is meant to.

WHY OPEN, AND WHY FROM BELOW

The spout is deep, narrow and blind. Filling it as a rising level leaves the air nowhere to hide; pouring into it from above traps a bubble under the stream, in the one part of the funnel that has to be a fluid boundary.

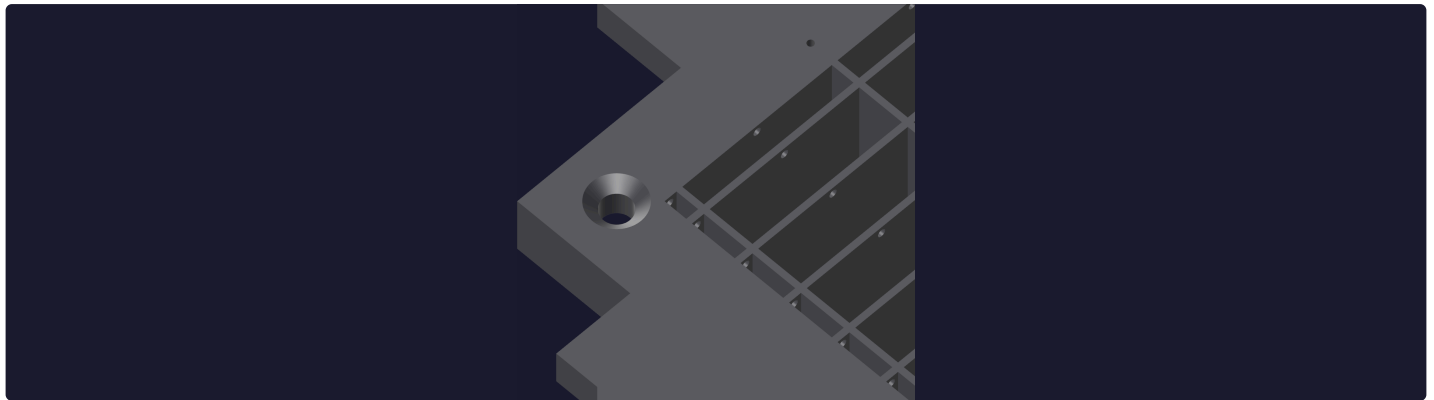
DONE WHEN

The whole batch is in the cavity, the spout pocket is full to the bowl floor, and no bubble is visible in it.

Lower the core

Slowly, straight down, and let the silicone find its own way out. Six ports in the brim are the only exits, and everything that leaves through them is air you are not casting in.

- 1 **Start it on the guides.** The four chamfered blades enter their bearings before the plug enters the cavity, so the core arrives square. Let it find them; do not push it sideways.
- 2 **Go down slowly and evenly.** The plug displaces the silicone ahead of itself; give it time to move rather than forcing a pressure wave through the spout.
- 3 **Watch the ports.** Air first, then silicone. A vent that never weeps is a vent that is blocked, or a corner that has not filled.
- 4 **Seat the skirt fully** — 10 mm of registration all round, down flat, no rock.
- 5 **Top up the dish.** Keep the casting full through the pour throat.



The corner of the brim ring carrying the 11 mm pour throat in its 20 mm dish. The five 2.5 mm vents take the other three corners and the two mid-sides, so no corner of the brim is unvented.

⚠ THE JACKS ARE NOT CLAMPS

Their tips must still be clear of the washers with the core seated. If a screw is bearing now, it is holding the mold *open*.

DONE WHEN

The skirt is seated flat all round, silicone has appeared at all six ports, and the dish is topped up.

Vacuum the filled mold

One more cycle, on the closed mold, to pull the air out of the corners the pour could not reach. It needs the ports open, room above the dish for foam, and your hand on the valve.

- 1 **Into the chamber on its shelf**, with the catch tray under it and the **backing-air exits still open**. Those channels are how chamber pressure gets inside the printed structure; covered, the mold fights the cycle.
- 2 **Pull down and watch the dish**. The silicone in the throat and the vents will rise and foam. That is the trapped air leaving.
- 3 **Admit air slowly**. A fast let-up drives foam back into the cast instead of collapsing it.
- 4 **Inspect the fill level and top up** through the pour throat while the silicone is still workable. Expect the level to have dropped.
- 5 **Repeat if the dish is still foaming**. Stop while you still have working time to top up afterwards.

⚠️ DEGASSED IS NOT THE SAME AS FULL

An underfilled mold does not become complete because it has been through the chamber. The top-up, done while the silicone still flows, is the step that finishes the part. After the working time it is a short shot and there is no recovery.

NOT A PRESSURE VESSEL

The printed tooling carries no pressure rating. The backing channels equalise it; they do not make it strong. If the halves visibly move, stop the cycle.

DONE WHEN

The dish has stopped foaming under vacuum, air is back in the chamber, and the casting is topped up to a full dish.

Hold it seated, and wait

The clock stops here. The only job left is keeping the core down, evenly, until the silicone is properly cured — and keeping the mold breathing while it does.

- 1 **Put the holding arrangement on.** Even pressure across the four arms, not a point load on the plate.
- 2 **Leave the backing-air exits uncovered.** All of them, on both halves, for the whole cure.
- 3 **Leave the pour dish and the five vents open.** Do not wipe them flush, do not cap them.
- 4 **Leave it alone.** Level bench, no draughts, room temperature, for the time the tin gives. Nudging a half-cured casting is how you get a witness line in the wall.

THE GUIDE'S CALL, NOT SPEC

No clamp, board, weight or force is specified anywhere in the source — only that something must hold the core seated, that the jacks do not, and that the air exits stay open. A flat board spanning the four arms with dead weight on it satisfies all three. Give the board a cut-out or a standoff over the pour dish so you can still see it.

⚠ WAIT FOR A FULL CURE, NOT A FIRM ONE

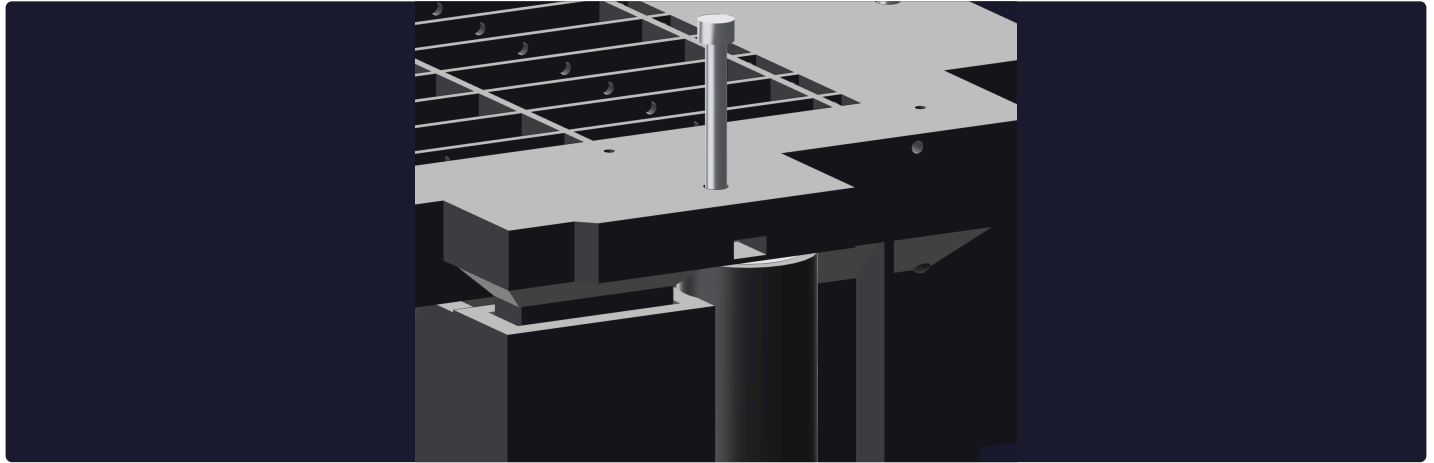
A part that is firm but not through-cured tears at the spout when you lift the core, and it will not survive the post-cure either. If in doubt, give it longer — nothing is lost by waiting.

DONE WHEN

The silicone in the pour dish is fully cured to the touch and the tin's stated cure time has elapsed.

Find zero

Before anything lifts, all four screws have to start from the same place. That place is the moment each tip just touches its washer.



One station: the screw standing through the core's **14 mm** lifting arm, and the solid **32 mm** column in the cavity below that carries the washer down to the bench.

- 1 **Take the holding weight and board off.**
- 2 **Turn each screw down by hand until its tip just contacts its washer.** By feel, with the hex key — you are looking for the first change in resistance, not for tightness.
- 3 **That is zero-lift.** All four now, before any of them go further.

🛑 THE HAND KEY, ONLY

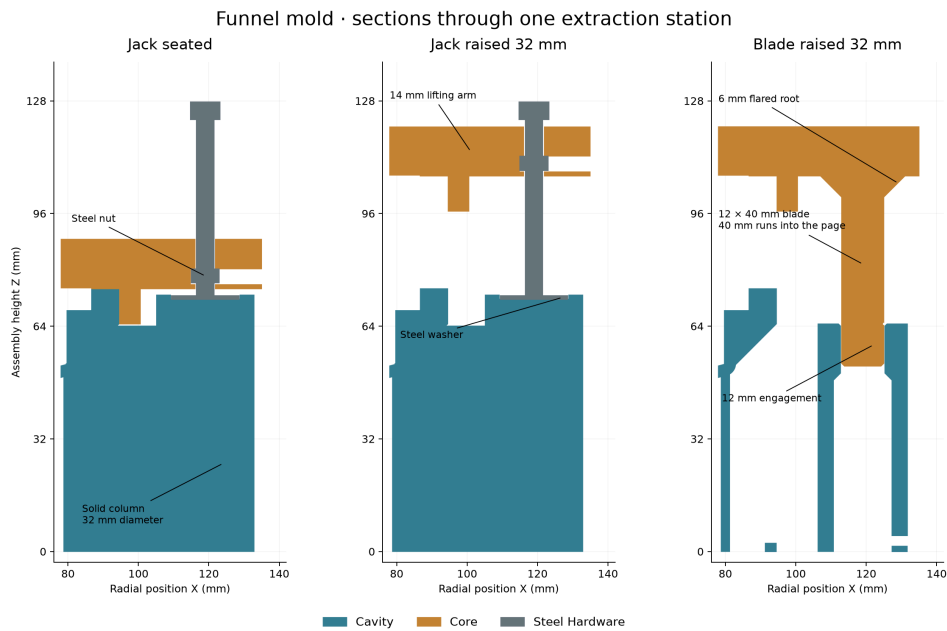
A powered driver will overload a printed ear long before a stuck casting lets go. The hex key that came with the screws is the tool, because resistance is the instrument you are reading.

DONE WHEN

All four screw tips are in contact with their washers and none has lifted the core.

Lift it 32 mm

Quarter turn, opposite-side pair, quarter turn, the other pair. Round and round. It is **forty full turns on every screw** and it is meant to feel long — you are separating a cured rubber part from a coated face without shocking either.



One station seated, and at full lift. The screw cannot push the washer anywhere, so the nut — and the core with it — rises instead.

WHAT YOU TURN	WHAT MOVES
One quarter turn, one screw	0.20 mm
One round of all four	0.20 mm of core
Forty full turns each	32 mm – the working lift

- 1 **Advance in opposite-side pairs**, a quarter turn each, then the other pair. Equal turns keep it level; they *do not* equalise force.
- 2 **Watch the gap at all four corners** as you go, and watch the rod.
- 3 **Stop immediately if** an ear flexes, a guide binds, or one side is running ahead. Find out why before you turn anything else.
- 4 **Stop at the travel marks.** Four shallow marks on the outside of the blades come level with the bearing mouths at exactly 32 mm. They are witnesses, not stops — nothing prevents you going past them.

DONE WHEN
The four travel marks are level with their bearing mouths and the core is still square.

Watch the rod, then lift it off

At 32 mm three things are true at once, and their order is the whole design: the rod is free of the core *before* the guides let go.

AT 32 MM OF LIFT		WHAT IT MEANS
Rod socket depth	28.8 mm	Already cleared. A stationary rod is out of the core.
Guide engagement left	12 mm	Still constrained square by all four blades.
Working lift	32 mm	The travel marks are at the bearing mouths.

- 1

Look at the rod before you lift further. If silicone has gripped it, the rod may have come up with the core instead of staying put. Ease it back down and free it by hand.
- 2

Then lift the core straight up, by hand, until the four blades clear their bearings. The screws come away in it, still in their nuts.
- 3

Set it down on its back, plug upward, somewhere it cannot rock on a blade.

⚠ DO NOT ROCK IT

A blade is far weaker to an in-and-out rock than to a sideways one, and by this point it is standing 54 mm free. Rocking is how you snap a guide root, and a core with three guides never comes off square again. If it will not come, put it back down and jack it further.

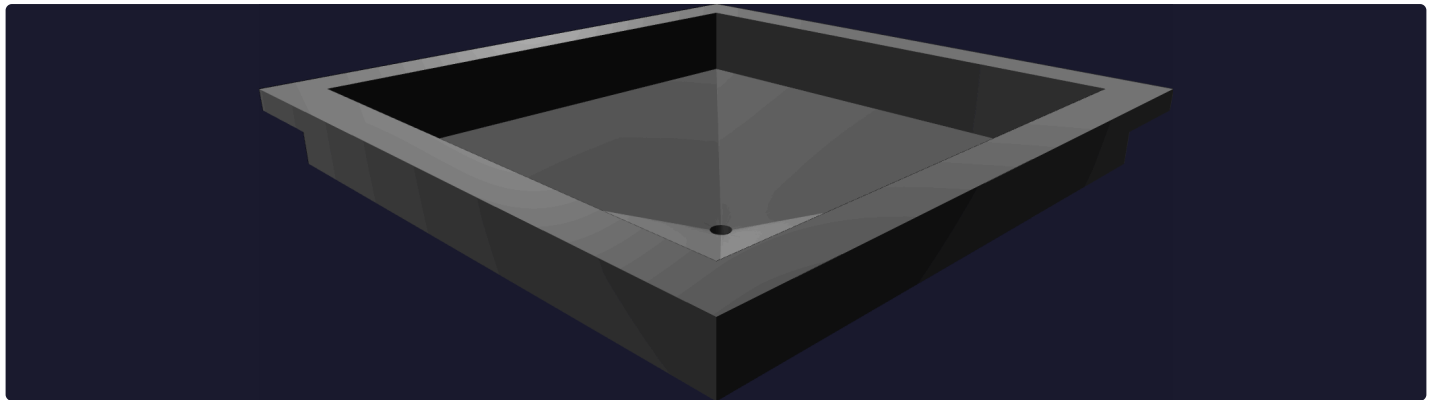
DONE WHEN

The core is off and set aside, the rod is still in the silicone, and all four blades are intact.

Peel it out, draw the rod

Two more movements, each on its own. The funnel comes out of the cavity first, with the rod still in it; the rod comes out of the funnel last.

- 1 **Start at the brim.** Lift one edge of the flange and break the seal all the way round before you pull anything upward. A 40A rubber wants to peel; it does not want to be dragged.
- 2 **Support the brim and the spout together.** The spout is the thin, deep part and it is still buried in the pocket — take its weight rather than letting the brim carry it.
- 3 **Lift the funnel clear** with the rod still through it.
- 4 **Draw the rod out axially.** Grip the exposed length — there is plenty of it — and pull straight. No twisting the funnel off it, and nothing that marks the rod's ground surface, because that surface is the next funnel's bore.



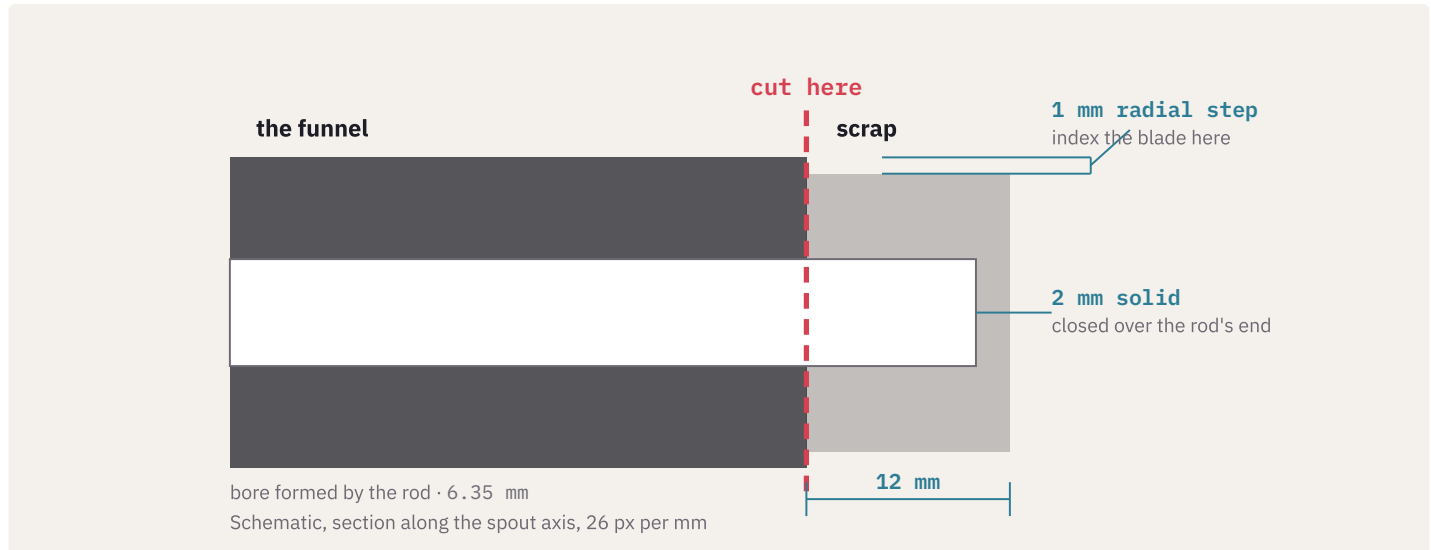
The finished shape: a brim you lift it by, a straight chute, and a floor that falls from every direction to one spout. What you are holding now still carries **12 mm** of sacrificial tip below that spout — page 15 cuts it off.

DONE WHEN

The funnel is out whole, the rod is out of it undamaged, and the cavity is empty and undamaged too.

Cut at the shoulder

The spout is cast **12 mm** too long on purpose, and the last **2 mm** of that is solid. One cut, at a step you can feel, opens the bore the rod already made.



Cut against the step and the finished bore is already ground to size by the dowel. Everything to the right of the dashed line is scrap.

- 1 **Find the shoulder by touch**, then cut square to the axis with a fresh, thin, very sharp blade in one pass. A sawing cut leaves a lip that fouls the drain stub.
- 2 **Trim the six pips** — one from the pour throat, five from the vents — flush with the brim, without gouging it.
- 3 **Check the bore.** Round, clean, unmarked, all the way through. This is the surface a push-fit collet will grip through the drain stub.

WHY THE TIP IS THERE AT ALL

A spout that ended at its finished length would have to be cast against a free rod end in open silicone, and its last millimetre would be the worst-formed part of the funnel. Casting 12 mm past the exit puts that region in the scrap, and leaves the cut face square on material that formed against a mold wall.

DONE WHEN

The spout ends square at the shoulder with an open 6.35 mm bore, and the brim carries no pips.

Wash, then bake

This funnel takes flavour concentrate and goes in the dishwasher. The room-temperature cure makes it a part; **the bake is what makes it a food-contact part**, because that is what drives off the volatiles and finishes the platinum cure.

- 1 **Wash the release film off.** Inside, outside, brim, spout and bore. Rinse and dry completely — water in the bore is water in the oven.
- 2 **Bake to the silicone maker's own post-cure schedule.** Its time and temperature are on the datasheet; they are not in this machine's files, and any figure you have seen elsewhere in the project is a planning estimate.
- 3 **Witness the temperature on the thermometer, not the dial.** Put the oven thermometer in the cavity and read it. That is what the shop oven is paired with it for.
- 4 **Cool it slowly and inspect it again.** A part that changed shape in the oven was not fully cured when it went in.

🚫 THE TOOLING NEVER GOES IN THE OVEN

The post-cure applies to the demolded silicone only. The PETG halves stay on the bench; they are unannealed, room-temperature tooling and heat will take their forming faces with it.

WHAT THE BAKE DOES AND DOES NOT SETTLE

It is the gate the wetted-surface screen is written around — residual platinum catalyst and cyclic siloxanes, on a coupon prepared the way production would be. That screen is what this project has. It is not a certification, and a coating or release coupon does not qualify the funnel on its own.

DONE WHEN

The funnel has been washed, dried, baked to the datasheet's schedule with the temperature witnessed on the thermometer, and cooled without distortion.

Inspect it, and read it

Four things to look for, and what each one is telling you about the run you just did.

WHAT YOU SEE	WHAT IT MEANS	NEXT TIME
Voids in the wall or the spout	Air that never left — entrained in the mix, stalled in the spout, or coming out of the print through a thin patch of coating.	Degas to a flat surface; pour the spout from below; check the coating for continuity.
Tacky or soft patches	Off ratio, unmixed material off the cup wall, or something on the mold face inhibiting a platinum cure.	Weigh to 0.1 g, scrape the cup, and re-run the coupon trial on the actual four materials.
Torn spout wall	Thin release, a lift that started before the rod was free, or the core rocked.	Re-release every pour; jack the full 32 mm before lifting by hand; never rock.
Short at the brim or a corner	Underfilled — not topped up after the vacuum cycle, or a blocked vent that never let its corner fill.	Top up while it still flows; clear all six ports before the next cast.

THE FIRST ONE IS A TEST PIECE

Measure this funnel against the nominal geometry — brim thickness, collar, the ramp transitions where the floor meets the walls. The first cast is the real dimensional and surface check on the tooling, and it is the only thing that can tell you the film came out even.

DONE WHEN

The funnel is free of voids and tack, whole at the spout, full at the brim, and measures to the nominal geometry.

Write the run down

The silicone's own numbers are not in this machine's files, and the ones that matter most are the ones you measured. Fill this in on the day, on this page, and the second cast starts from something better than memory.

CAST	VALUE	CAST	VALUE
Date		Working time used	
A / B mass		Cure to demold	
Pigment mass		Post-cure °C / h	
Degas — time to flat		Witnessed temp	
Vacuum cycles		Top-up after vacuum	
Ports that ran silicone	/ 6	Result	

Before the next one

- ☐ Both forming faces washed clean of cured silicone and re-released.
- ☐ All six ports and every backing-air channel clear again.
- ☐ Washers checked for a scored bearing track; replace a deeply scored one.
- ☐ Screw tips still flat and burr-free.
- ☐ Rod's ground surface unmarked.
- ☐ Coating inspected on both faces — re-sand, re-seal and re-release as the film wears.

The funnel's drain stub and worm clamp are a separate joint, made after this and washed with the part — card SA-06 owns it, and this guide stops here.